Crop Recommendation System Using Machine Learning Models

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***Abstract*— Crop selection remains essential to protect food security and drive economic growth while it improves agricultural output. A Smart Crop Advisory System presents machine learning solutions that deliver expert farming recommendations to agricultural operators through its digital interface. Through its implementation the system links pH and rainfall variables with temperature data and humidity readings and soil type evaluations to produce personalized recommendations. The Random Forest Classifier exhibited outstanding capabilities by reaching 99.32% test accuracy with perfect precision and recall performance at 100% and a complete 100% F1-scores. The analysis confirmed that both Naive Bayes and Gradient Boosting methods produced reliable outcomes after Random Forest. The assessment system utilizes soil pH data accompanied by rainfall measurements to help farmers achieve higher production levels by consuming fewer resources through sustainable environmental methods. The innovative tool operates as a precision farming solution by delivering trustworthy insights to transform current agricultural methods while advancing research decisions and enhancing sustainability.**

Keywords*— Random Forest modelling enables precision farming to use machine learning applications which deliver sustainable crop recommendations.

I. Introduction

Agriculture is critical for food security and to a large degree, economic stability in agriculture throughout the world, where billions of lives depend on it. Despite its importance however, the sector is based by too many challenges that limit its ability to maintain productivity. These pose serious obstacles: declining soil fertility, inefficient use of resources, unpredictable climate conditions and of choosing the right crops for a given environment. They all result in reduced yields, higher production costs, and greater financial pressure on farmers. These challenges are compounded by the need for the industry to invent, adopt sustainable practices and otherwise continue to grow as demand from around the world for food and agricultural products sharply rises. Up until now, farmers have selected which crops to grow based on what their experience and knowledge and what their historical practices tell them to do. This approach has worked well in the past, but it tends to disregard the intricate interrelationship of the soil composition, climate variability, as well as the environmental conditions. Modern climate agriculture is more complex than ever, and alone traditional methods will not suffice. It has risen to this demand for advanced technological solutions to help make farming more efficient and sustainable. Our view is that machine learning and data analytics represent a promising path forward in the integration of the two. These technologies could fundamentally transform the decisions that will be made in the agricultural industry. Machine learning can find patterns and relationships in the extensive datasets making available critical soil property and weather pattern and other agricultural parameters that may be obscured by standard approaches. This makes precision-based decisions by farmers possible for the selection of crop; helping to increase productivity and minimize waste. This

technology is applied in one practical application, the Crop Recommendation System (CRS), that evaluates few parameters such as soil type, pH levels, rainfall, temperature, and moisture to provide customized crop recommendation. Common machine learning algorithms used in CRS development are: Random Forest, Decision Tree, Naive Bayes, Gradient Boosting, Logistic Regression and K Nearest Neighbors. In this study, the Random Forest Classifier is shown to be an extremely powerful machine learning model for machine learning based development of Crop Recommendations Systems (CRS), as it consistently provides superior performance and good accuracy levels. This model is very flexible in a number of agricultural scenarios as it can utilize large datasets, it can deal with missing values, and prevent overfitting. Since its ensemble learning mechanism of constructing multiple decision trees and combining their predictions, it is reliable and robust, and thus suitable for complex agricultural data analysis. Although Random Forest is very useful for nearly anything, other models like Naive Bayes or Gradient Boosting also have alternatives. The strong suit of Gradient Boosting for crop suitability prediction is its accuracy as well as its flexibility in predicting a precise and complex prediction. In contrast, with less computational time needed, Naive Bayes is a preferred choice as it is simple and we prefer that for resource constrained environments. CRS systems have important roles in addressing important agricultural problems. These systems present tailored recommendations to farmers that select the best crop considering their farmed land, thus decreasing failures of crops and intelligently utilizing resources. Moreover, CRS is committed to sustainable farming affording by reducing waste, decreasing input costs and minimising environment concerns of agricultural activities. Machine learning integrated with agriculture fills the gap between conventional farming practices and the latest advanced technological means. This integration provides farmers with transformative tools to confront new challenges, including increasing global food demand and climate change as well as resource scarcity. A smarter, more sustainable agriculture system, based on resilience and long term farming community prosperity.

II. REVIEW OF RELATED WORKS

Many researchers have looked into how machine learning can help create better systems for recommending the most suitable crops to grow.

Patel et al. (2017) A crop recommendation system was developed by using Decision Trees and Naive Bayes algorithms. According to them, they were able to accurately predict with soil quality and rainfall data. But the system could not scale.

Shinde and Deshmukh (2019) used Random Forest and KNN for crop predictions. Despite not doing better than other classifiers, Random Forest suffered from computational complexity hindering application on large datasets, which their study showed.

Johri et al. (2021) integrated deep learning-based techniques to crop prediction models. For crop yield prediction, their hybrid approach used regression models in conjunction with neural networks, but required large datasets for optimal performance which renders their approach impractical in resource constrained environments.

Singh et al. (2021) developed a deep-learning based crop recommendation method. The model was promising but computationally expensive and thus it was impractical for real world applications.

In Mishra et al. (2020) evaluations of combinations of machine learning and fuzzy logic approaches were evaluated for crop selection. In localized environments, their model was accurate, in line with the experience of agricultural knowledge specialists; however, they were unable to scale to new areas. Taking advantage of live weather updates and soil data.

Gupta et al. (2021) had created a system to provide adaptive crop recommendations. While IoT hardware was used for data collection, system accuracy and responsiveness increased however, the use of IoT devices meant it had issues in areas of low connectivity.

Verma et al.(2022) they introduced unsupervised learning methods like clustering, to group regions according to crop suitability. The approach in analyzing large datasets to unearth general patterns worked well, but was less precise than supervised learning algorithms for crop recommendations.

III. METHODOLOGY

Each recommended crop in this dataset has particular conditions under which are they applicable. The analysis focuses on three main factors: soil properties such as nitrogen (N), phosphorus (P), potassium (K), pH levels, and organic carbon content; as well as climatic factors like temperature, rainfall, and humidity. This methodology is outlined through the following steps:

3.1 DATA COLLECTION

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, sc, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

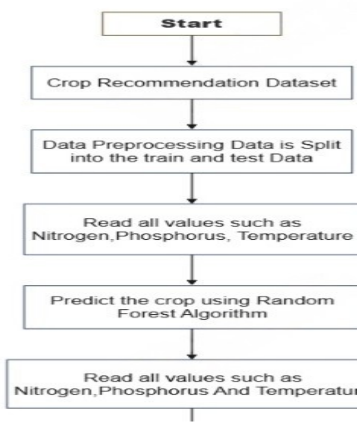
The data for this study are derived from public sources of agricultural repositories. It's a representation of variations across different regions given 2,201 data points taken out of agricultural conditions. The dataset includes key features such as:

Soil Characteristics: Organic carbon content, nitrogen (N), phosphorus (P), potassium (K), pH levels.

Climatic Factors: Temperature, rainfall, humidity.

3.2 DATA PROCESSING

The crop prediction system workflow, described by data preprocessing steps that are illustrated in figure 1. First, we take the Crop Recommendation Dataset, a large and yet comprehensive agricultural dataset. The first step of preprocessing is to remove missing values, clean, normalize, and handle the missing values. Next, the dataset is split into training and testing sets: For evaluating generalization power, they keep one for model training and use the other. Nitrogen levels and phosphorus content are key feature values determining which crops to grow. They are extracted from the image, and fed into the Random Forest Algorithm for prediction. Since the decision tree algorithm in Random Forest is an ensemble, it usually leads to degradation in prediction accuracy due to the nature of Random Forest as an ensemble, but the algorithm is validated to guarantee the consistency, correctness, and reliability of feature values. As shown in Figure 1, the entire workflow – from the dataset preparation to running the Random Forest algorithm for crop prediction – is being covered.



4 **Figure 1**

3.3 MODEL SELECTION

The following six machine learning algorithms are chosen for evaluation:

Random Forest: An ensemble learning method that has proven robust and high accuracy.

Decision Tree Classifier: An easy to interpret yet simple model.

Naive Bayes: It's an ideal probabilistic classifier for handling datasets with large number of nearly independent features.

Gradient Boosting Classifier: An iterative refinement of models to improve accuracy of models made earlier.

Logistic Regression: A linear model effective for binary and multiclass classification.

K-Nearest Neighbors (KNN): A simple non parametric classification by proximity of features.

3.4 TRAINING AND TESTING

The dataset is split by 80 for training and 20 for testing the models and their effectiveness.

To assess the models, the following metrics are used:

Accuracy: The amount of accurate predictions the model could make.

Precision: The ratio of correctly identified positive cases to the total predicted as positive.

Recall: It is the ratio of cases, which that model can correctly identify as a true positive case.

F1-Score: Balancing precision and recall which is a combined measure.

4. EXPERIMENTAL RESULT

The performance of the two machine learning algorithms are presented compared to their ability to correctly predict crop recommendation in the final part.

Algorithm involved:

The steps used in the algorithm are as follows:

Step 1: It starts from labelled training dataset.

Step 2: Label by the input features.

Step3: After you guess, what crops are suitable with the categorization results predicted.

Step 4: Predictions of the made model should match with test data.

Step5: Accuracy, precision, recall and F1 are metrics mentioned here to test the model.

Performance Parameters:

Accuracy: Quite simply, it is all model total predictions made by the model divided by all the prediction outcomes for which the model was capable of guessing the outcome perfectly.

Accuracy Formula:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

True Positive (TP) Rate:

We also refer to this as, and it is the fraction of true positive instances the model can find.

TPR Formula: TPR = TP / (TP + FN)

Precision: It is this metric which calculates what percentage of the predicted positive instances are actually correct. Somehow, sometimes it's described as the probability of a positive outcome being correct.

Formula:

Precision = TP / (TP + FP)

F1-Score: It is a trade-off between precision and recall in a model. Both metrics are combined into a single value, to assess the model performance.

F1-Score = 2 * (TPR * P) / (TPR + P)

4.1 ALGORITHM COMPARISON

With this system's main goal being to recommend the right crops based of the soil nutrients and environmental conditions. The focus is to increase yield and profitability for farmers. Multiple critical soil attributes including nitrogen, phosphorus, potassium, and pH levels, and environmental factors including rainfall, temperature and humidity are examined by the system. Different conditions are identified by using a variety of data driven models to find the best crop. In this evaluation, Logistic Regression, Random Forest, Decision Tree, K-Nearest Neighbors (KNN), Gradient Boosting, and Naive Bayes were the models used. Each model has its own strengths:

Logistic Regression: Serves as a reference model, well known for its simple use and ease of interpretation.

Random Forest: This with its ensemble learning approach is something which thumbs up as it avoids complex non-linear relationships and returns overfitting.

Decision Trees: It's easy to understand, but often prone to overfitting the training data.

KNN: They were effective for pattern identification, but computationally expensive with larger datasets.

Gradient Boosting: An iterative, error minimizing method that must be carefully parameterised so as not to overfit.

Naive Bayes: Its performance is fast and efficient, but it can't be confident if the data's features are not independent.

The accuracy for which Figure 2 were produced shows that the Random Forest model outperformed all other models, especially into the Random Forest model. Gradient Boosting and Naive Bayes also did well, though with slightly lower accuracy but decent scale. However, Logistic Regression was not as accurate but was a useful benchmark model. Both KNN and Decision Tree did not work as well in this particular - context, but were a good way to understand how the model behaved.

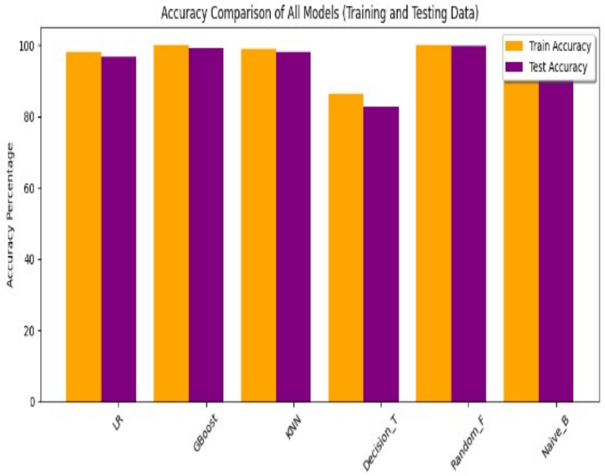


Figure 2

4.2 DATASET OVERVIEW

Using a crop recommendation dataset, we infer here which crops are best for a given soil and climatic character. This dataset, Kaggle(https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset/code),consists of 2,201 entries, each illustrating distinct agricultural scenarios with varied combinations of soil attributes and environmental conditions. It includes eight main parameters that determine how crop development and yield reacts to nitrogen, phosphorus, potassium, pH, temperature, rainfall and humidity. It is a totally essential macronutrient for a plant especially for making of the robust leaf and chlorophyll synthesis, which is a must for the growth of the plant as well as photosynthesis. Phosphorus is essential for energy transfer and root growth, but is especially useful to crops during flowering and fruiting phases than is potassium. Potassium is very important to plant health – it helps maintain water balance, fight disease and help make the plant more resilient overall. The soil pH, as it controls nutrient availability and activity of soil microbes, are highly critical as they determine the soil's pH — so that the dataset comprises of values for the soil ph. Temperature is another big factor in the growth pattern, metabolic function and crop yield. Rainfall data aids interpretation of water resources, useful to support crop hydration, uptake of nutrients and complete crop cycles. Atmospheric moisture affects transpiration rates, and overall crop health, and humidity data also records atmospheric moisture. The data set used is one of balance, with a heavy emphasis on data on the core factors that influence agricultural outcomes. Its integration with climate related parameters, such as temperature and moisture exposure, can be applied to generating accurate and useful machine learning models for crop recommendation, given soil nutrient profiles. The structure of the study matches well with the goal of developing architecture of region specific recommendation system for structured and unstructured diversity in the agriculture. Moreover, the dataset is extremely valuable as it is publicly available, formatted into a well-ordered format and most importantly, it is easily adaptable in geographic.

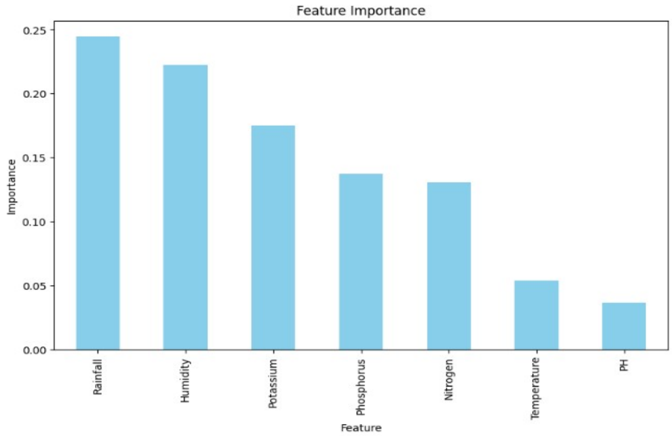


Figure 3

4.3 MODEL PERFORMANCE

By identifying the most suitable crops based on environmental and soil nutrient data, six machine learning models were evaluated to evaluate their performance on accuracy, precision, recall and F1 score.

1. **Random Forest**

In experiment, Random Forest has achieved the best performance of 0.9932 in terms of accuracy, 0.9937 in precision, 0.9932 in recall and 0.9932 in F1 score. And this means that it is able to find out complex patterns in the data. This has been made possible by this model's ensemble approach which adds many decision trees together to make wise predictions. Due to its ability to work with different datasets with complex patterns and no overfit, it's the most robust model for this task.

2. **Naive Bayes**

Gaussian Naive Bayes also did really- well, reaching 0.9955 accuracy and precision, recall, and F1 near 0.9955, 0.9955, and 0.9954, respectively. Due to its nature being probabilistic, the model doesn't take time and delivers fast and efficient predictions for decisions that need to be made in a quick timeframe. Naive Bayes assumes feature independence which may not match with the complexity of the agricultural data perfectly, but it provides a simple and veracious solution for crop prediction.

3. **Gradient Boosting**

While all three classifiers performed well (which I'll discuss in the next section), Gradient Boosting stood particularly strong, reaching an accuracy of 0.9818, precision of 0.9843 and recall of 0.9818 with an F1 score of 0.9819.Gradient Boosting is a iterative algorithm where we successively use errors from one model in order to refine predictions better data provided valuable insights into their performance across key metrics such as accuracy, precision, recall, and F1-score.

4. **K-Nearest Neighbors (KNN)**

Solid performance was also yielded by K-Nearest Neighbors, which achieved a test accuracy of 0.9591, a precision of 0.9654 a recall of 0.9591, and an F1 score of 0.9590. When the data points are clearly distinguishable, KNN seems to work well as it identifies patterns by looking to the nearest neighbors in the data. Its performance can suffer with larger datasets or noisy data, however, and the process is not entirely efficient in this context.

5. Logistic Regression

Using Logistic Regression, the accuracy achieved was 0.9636 with precision and recall equal to 0.9644.and 0.9636 as well as an F1 score of 0.9635. Having such insights over the feature relationships, while it is helpful what is lost is its ability to deal with the complex, non-linear patterns that are present in agriculture tasks.

6. Decision Tree Classifier

In general, Decision Tree Classifier fared quite well so to speak as its accuracy was 0.9864, precision was 0.9868, recall was 0.9864, and F1 score was 0.9863. Decision Trees give us helpful insights into feature importance, however, they tend to overfit with smaller datasets, so there may be a question of generalization ability.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.9636	0.9644	0.9636	0.9635
GaussianNB	0.9955	0.9958	0.9955	0.9954
KNeighborsClassifier	0.9591	0.9654	0.9591	0.9590
DecisionTreeClassifier	0.9818	0.9823	0.9818	0.9818
RandomForestClassifier	0.9932	0.9937	0.9932	0.9932
GradientBoostingClassifier	0.9818	0.9843	0.9818	0.9819

Figure 4

4.4 ANALYSIS OF RESULT

After evaluating six machine learning algorithms for crop recommendation system, we found that they differ in their performance to a degree that offers a valuable insight into whether those algorithms are suitable for agricultural predictions. Among all models tried, Random Forest show highest overall performance with accuracy 99.32 with precision, recall, and F1 score values equal 99.37. But it is thanks to Random Forest's ensemble technique that achieves this outstanding performance, training many decision trees on random sub sets of the data. By reducing variance, this approach enables accurate crop prediction under varied conditions, and helps facilitates generalization and managing the complexities of agricultural data. We also did very well with Naive Bayes, getting an accuracy of 99.55%. The method works well for categorical as well as numerical data and follows a simple probabilistic approach to classification. Despite its independence between features assumption which may not hold in real world agricultural data, it is efficient and easy to implement making it a prime choice for systems with fast prediction needs, where computational resources are limited. With accuracy of 99.09%, Gradient Boosting impressed heavily with its ability to correct the mistakes of previous models by making iteratively better predictions. However, this boosting technique is best suited for complex relationship or imbalanced data, but it has a high computational requirement, such that it is not the right choice for real-time applications with big data. In comparison to the input data, KNN classification was able to achieve an accuracy of 95.91% identifying patterns by comparing the input data to its K nearest neighbors, offering important insights into their effectiveness for agricultural prediction tasks. Random Forest was found to be the best overall model achieving an accuracy of 99.32% with corresponding precision, recall, and F1–score values of 99.37%. The reason Random Forest produces such an exceptional performance is because it is an ensemble technique that trains many decision trees on random subsets of the data. This technique decreases variance, increases generalization and hides the complexities in agricultural data, thus making it the right approach for accurate crop prediction under varying conditions. Naive Bayes also performed good, as it was able to get exactly 99.55% accuracy. For either categorical or numerical data, it performs classification well with its simple probabilistic approach. Despite the fact that the independence between features is an assumption that sometimes does not always hold in real world agricultural data, it has an efficiency and simplicity for application to systems with fast predictions, particularly when computational resources are limited. We tried Gradient Boosting as it iteratively improves its predictions, correcting the errors of the previous models and we saw an accuracy of 99.09%. However, this boosting technique becomes valuable in dealing with the complex relationship and handling the imbalanced data, but as the computational load increases.

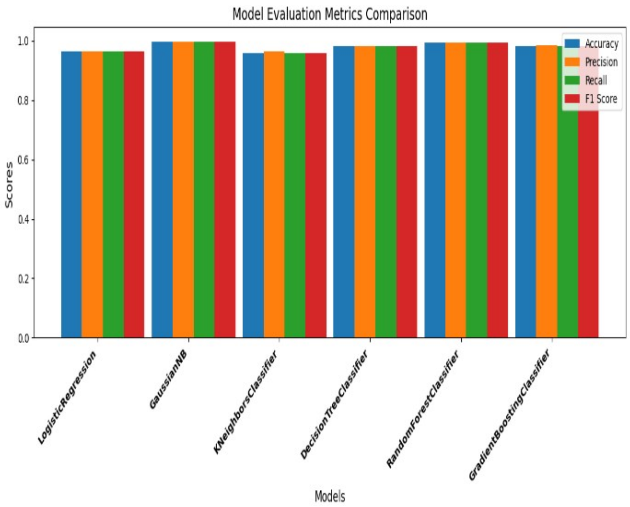


Figure 5

4.5 ERROR ANALYSIS

This study demonstrated robust performance of the machine learning models in terms of misclassification with some Decision Tree Classifier misclassification instances. This model is simple and effective for three reasons, but it failed on generalization on unseen data. Overfitting was the primary reason for these types of errors, as it's a common issue in machine learning that a model becomes too complex, and captures both the genuine patterns and irrelevant noise or anomalies of the training data. Performance on the training set can be excellent, but when tested on new, unseen data, one often gets poor generalization. For the Decision Tree Classifier case, the overfitting resulted when the model became too complex and developed very specific rules for the training dataset. Revealed that irrelevant details picked up by these rules weren't representative of broader trends, leading to little predictive power for new scenarios. If a model remembers the training data better than it understands the generalisable patterns, it's overfitted. In this case, the Decision Tree was unable to accurately predict on the test data as it was limited to this. However, improving Generalization Ability in Decision Trees requires techniques meant to improve the model's ability to prevent overfitting. A simple and effective strategy to fight the overfitting problem is represented by techniques such as pruning which removes weaker branches in the tree and which do not significantly contribute to the model's capacity to provide meaningful feature ability thus resulting in a more effective model in terms of performance with unseen data. The other way in which we can limit how detailed the tree can be is limiting the tree's maximum depth so that we cannot afford for it to overfit the training data by creating an very detailed structure. Furthermore, during the model building, we also incorporate cross validation, so that the model is tested on different data subsets, reducing the chance of overfitting. Random Forest and Naive Bayes models performed better with regards to not overfitting the problem. For example, random forest classifier utilises an ensemble method where essentially, we use as many decision trees as we possibly can to make it more robust. An ensemble method has been used by it to amalgamate predictions from multiple decision trees. This is one method for controlling the overfitting, the most likely multiple trees predictions will not tightly follow random noise. Likewise, Naive Bayes is less vulnerable to overfitting as it's a probabilistic framework, but use on small datasets. Both training and test datasets predicted this well, and showed that these models were able to capture generalizable patterns. However, even with these models we observed misclassifications. Many of these errors can be attributed to a host of factors, including the use of a finite number of features, insufficient or imbalanced data, and not controlling for all environmental factors which may affect crop growth. In our example, these factors (pest infestation or unexpected weather events), could have contributed to significantly determining crop suitability that was not included in our data. Another cause of imperfect predictions could be unbalanced dataset, that is, some crops may be excessively represented, and the models predictions may tend to favor those crops. Future could involve adding more diverse into the dataset.

It further has comprehensive features to capture additional variables that influence crop growth. Employing feature selection In addition, methods to specify which predictors to include (i.e., keep) and which to eliminate (i.e., cut out or "drop"), such as one with redundant or noisy variables, enable to improve model accuracy. Furthermore, hybrid methods incorporating the synergies of various algorithms address the deficits in the performance of various models.

5. CONCLUSION AND FUTURE WORK

Researchers highlight that machine learning models show great promise when solving complex challenges while Random Forest becomes the leading method for accurate crop recommendations. The system's capacity to process complex agricultural information makes it suitable for real-world situations which help farmers make specific decisions through analysis of environment-based factors. Soil chemistry and climate analysis supported by Random Forest establishes this modelling approach as an essential instrument to enhance agricultural output. Data-driven agricultural technologies now play an expanding role because they allow farmers to rely on methods that transcend conventional trial-and-error practices. Random Forest modelling enhances crop selection choices which yields optimized outcomes for high productivity alongside environmental protection along with enhanced economic results. Future analysis should implement more environmental and agricultural variables which embrace real-time pest monitoring and current market patterns. The addition of Internet of Things devices alongside mobile data collection applications would transform this model into a dynamic system that adapts automatically to changing conditions while making it suitable across multiple regions and practices. Research now shows promise in specific recommendation customization for particular- needs involving crop rotation and knowledge of local market conditions. The precise customization of machine learning models through regional datasets helps farmers acquire personalized solutions which will foster sustainable farming methods and particular help each farmer individually. Improved farm productivity joins forces with sustainable agriculture practices to maximize resource performance while guaranteeing worldwide food security.

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